

SAN BERNARDINO, Switzerland

WMO: 067830

Lat: 46.4636N

Lon: 9.1847E

Elev: 1638

StdP: 83.13

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -13.6	(c) -11.8	(d) -23.4	(e) 0.6	(f) -8.2	(g) -20.9	(h) 0.7	(i) -7.8	(j) 11.7	(k) -5.9	(l) 10.4	(m) -5.1	(n) 2.6	(o) 320	(p) 0.668

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.6	22.0	14.0	20.4	13.7	19.0	13.0	15.7	19.5	14.8	18.6	13.9	17.5	2.8	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.2	12.4	17.6	13.3	11.6	16.5	12.4	11.0	15.6	50.4	19.6	47.3	18.8	44.5	17.6	19.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 9.2	(b) 8.0	(c) 7.1	(e) -16.4	(f) 24.2	(g) 2.3	(h) 1.9	(i) -18.0	(j) 25.6	(k) -19.4	(l) 26.7	(m) -20.6	(n) 27.8	(o) -22.3	(p) 29.1		
			DB	WB	WB	WB	WB	WB	WB	WB	WB	WB	WB	WB		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.6	-3.5	-3.1	-0.2	2.9	7.2	11.5	13.3	13.0	9.2	5.6	0.7	-2.3
	DBStd	6.95	3.79	4.24	3.83	3.28	3.12	3.29	2.65	2.70	2.95	3.37	3.83	3.87
	HDD10.0	2296	418	367	316	214	98	21	6	6	49	139	279	382
	HDD18.3	5026	677	601	574	464	346	204	155	167	274	394	529	641
	CDD10.0	314	0	0	0	1	10	68	109	98	24	3	1	0
	CDD18.3	2	0	0	0	0	0	1	0	1	0	0	0	0
	CDH23.3	13	0	0	0	0	0	4	2	7	0	0	0	0
	CDH26.7	1	0	0	0	0	0	1	0	0	0	0	0	0
(14) Wind	WSAvg	2.4	2.8	3.1	2.8	2.5	2.3	2.3	2.4	2.1	2.1	1.9	2.2	2.5
(15) Precipitation	PrecAvg	1533	68	57	69	124	159	168	157	171	159	157	170	73
	PrecMax	2151	200	212	235	351	459	394	283	292	377	521	696	194
	PrecMin	855	7	7	8	22	43	39	51	27	31	17	13	2
	PrecStd	324	51	46	52	79	90	83	62	59	110	107	154	48
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.6	10.0	11.3	15.0	19.1	23.4	23.2	24.2	20.0	17.0	13.1	9.2
		MCWB	2.0	2.5	3.7	7.4	11.3	14.2	14.9	14.5	11.9	8.8	5.5	1.7
	2%	DB	5.6	7.1	9.3	12.2	16.8	21.3	21.6	21.8	17.8	14.3	9.7	6.3
		MCWB	0.4	1.2	3.0	5.7	9.6	13.7	14.1	14.2	11.6	7.7	4.4	0.8
	5%	DB	3.7	5.0	7.3	10.1	14.8	19.5	20.2	19.8	16.0	12.4	7.4	4.3
		MCWB	-0.6	0.1	2.0	4.7	8.8	13.0	13.8	13.7	10.9	7.2	3.0	0.0
	10%	DB	1.7	2.9	5.5	8.1	12.7	17.8	18.8	18.2	14.2	10.6	5.5	2.7
		MCWB	-1.5	-1.3	1.1	3.6	8.0	12.2	13.2	13.1	10.1	7.0	2.7	-0.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.1	3.2	4.7	8.1	12.2	16.6	17.2	16.7	14.1	10.6	7.0	3.4
		MCDB	7.4	8.2	9.3	13.4	16.9	20.5	20.9	20.8	17.5	13.8	10.8	6.5
	2%	WB	1.3	1.6	3.5	6.5	10.8	15.0	15.8	15.5	12.9	9.5	5.5	2.0
		MCDB	4.2	6.1	7.3	10.7	15.0	19.3	19.5	19.4	15.9	11.7	8.0	4.3
	5%	WB	0.1	0.4	2.6	5.4	9.6	13.8	14.8	14.6	11.7	8.7	4.4	0.9
		MCDB	2.1	4.0	6.2	8.8	13.4	18.1	18.4	18.2	14.6	10.7	6.2	2.9
	10%	WB	-0.8	-0.6	1.7	4.2	8.5	12.7	13.9	13.6	10.8	7.8	3.4	0.1
		MCDB	1.2	2.2	5.0	7.2	11.7	16.8	17.4	16.9	13.3	9.8	5.0	1.9
(35) Mean Daily Temperature Range	MDBR		6.9	7.1	7.2	7.1	7.7	8.5	8.6	8.2	7.6	7.2	6.0	6.6
	5% DB	MCDBR	9.6	9.5	9.7	10.6	11.2	11.3	10.8	10.8	10.5	10.8	9.7	9.1
		MCWBR	6.1	5.8	5.5	5.7	6.2	6.1	5.8	5.9	5.9	6.1	5.8	5.7
	5% WB	MCDBR	7.6	8.4	8.3	9.2	9.9	10.3	10.1	9.8	8.9	7.9	7.3	6.9
		MCWBR	5.3	5.5	5.1	5.5	6.0	6.3	6.2	6.0	5.6	5.2	5.0	5.0
(40) Clear-Sky Solar Irradiance	taub		0.240	0.264	0.307	0.353	0.367	0.377	0.370	0.362	0.342	0.316	0.272	0.237
	taud		2.332	2.268	2.203	2.184	2.280	2.300	2.319	2.358	2.399	2.448	2.445	2.368
	Ebn at Noon		877	915	916	900	898	888	888	882	871	845	830	841
	Edh at Noon		72	96	121	136	128	127	123	113	99	81	66	62
(44) All-Sky Solar Radiation	RadAvg		1.61	2.58	3.79	4.47	4.80	5.64	5.64	4.89	3.81	2.52	1.57	1.34
	RadStd		0.20	0.30	0.55	0.72	0.48	0.50	0.48	0.43	0.37	0.30	0.24	0.17

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only		+0.39	N/A	N/A	+0.72	+0.56	+0.42	N/A	-151	N/A	N/A
(47) Regional (117 neighbors)		N/A	N/A	N/A	+0.72	+0.42	+0.32	N/A	N/A	+60	+17

Nomenclature: See separate page